Action Recognition of Persons in Aerial Videos

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MTP Final Presentation

Topics to be Covered

- Introduction & Problem Statement
- Dataset Overview & Visualization
- Paper Approach
- Full Pipeline
- Pipeline for Models
- Proposed Framework
  - YOLOv8-based Detection
  - Hybrid Tracking Strategy
  - TimeSformer for Action Recognition
- Results and Performance Analysis
- Conclusions

Introduction & Problem Statement

Problem Statement

- Aerial videos introduce unique challenges:
  - Very small human appearance (high altitude)
  - Drone motion and camera instability
  - Multiple people in complex scenes
  - Subtle and overlapping actions
- **Key Problem:**
  - How to reliably detect, track, and recognize actions for multiple people in aerial videos?

Key Idea

Person-Centric Pipeline

Detect $\rightarrow$ Track $\rightarrow$ Recognize

- Analyze each person individually instead of the whole frame

"Detection tells where, tracking tells who, and TimeSformer tells what."

Dataset Overview & Visualization

Dataset: Okutama-Action

- UAV dataset captured using **DJI Phantom 4**
- **Key Characteristics:**
  - 43 minutes of aerial video footage
  - 4K Resolution: 3840×2160
  - 30 FPS (high temporal consistency)
  - Drone altitude: 5–10 meters
- **Human Action Classes (12):**
  - Calling, Carrying, Drinking, Handshaking
  - Hugging, Lying, Pushing/Pulling, Reading
  - Running, Sitting, Standing, Walking
- Frames extracted and annotations processed for training

Challenges in Okutama Dataset

- **Small human size** due to aerial view
- **Drone motion** causes camera instability
- **Multiple people** in a single frame
- **Occlusions and overlaps**
- **Class imbalance** (some actions are rare)
- Difficult to distinguish **subtle actions**

Class Distribution in Okutama Dataset

Important Note:

- Dataset is **multi-label** and **video-based**
- Exact per-class sample counts are not explicitly defined

Observed Distribution:

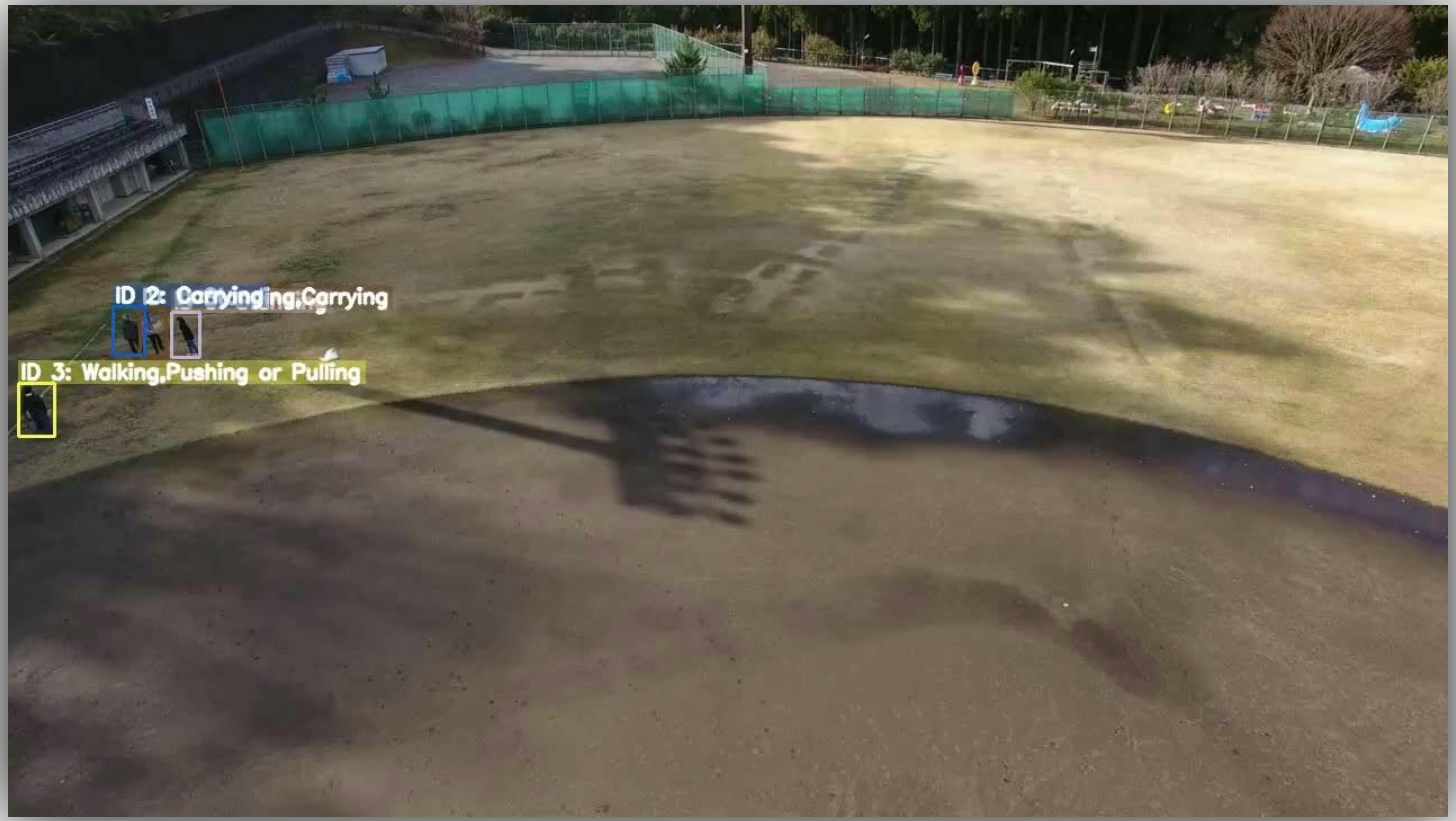
- **Frequent Classes:** Standing, Walking, Running
- **Moderate Classes:** Carrying, Sitting, Pushing/Pulling
- **Rare Classes:** Reading, Calling, Drinking, Hugging, Handshaking, Lying

Key Insight:

- Dataset is **highly imbalanced**
- Leads to **higher Micro F1** and **lower Macro F1**

Real Dataset Visualization

Real Dataset Visualization



Original Paper Approach (Okutama-Action)

- Based on **SSD (Single Shot Detector)**
- Adapted for **action detection in aerial videos**
- Pipeline:
  - Frame-level detection using SSD
  - RGB + Optical Flow inputs
  - Fusion of spatial and motion information
  - Linking detections into action tubes

Pedestrian Detection using SSD

- Model: SSD with VGG backbone
- Input size: 512×512
- **Performance:**
  - mAP@0.5 = **72.3%**
- **Observation:**
  - Performance drops for small objects (high altitude)
- **Insight:**
  - Detection is relatively easy compared to action recognition

Action Detection Framework

- Extend SSD to classify **actions** instead of only detecting objects

Three-Stage Pipeline

1. **Frame-level Detection**
   - SSD predicts bounding boxes + action labels per frame
   - Uses:
     - RGB images (appearance)
     - Optical Flow (motion)
2. **Fusion of Streams**
   - Combine RGB + Optical Flow outputs
   - Captures spatial + temporal information
3. **Action Tube Generation**
   - Link detections across frames
   - Ensures temporal consistency

Note

- Paper focuses evaluation on **frame-level SSD detection**

Action Detection Performance

Input Image	mAP (%)
RGB 512×512	15.39
RGB 960×540	18.80
Optical Flow 512×512	6.47

- Action detection is significantly harder than detection
- Temporal actions are difficult to distinguish
- Higher resolution improves performance

Key Gap: Detection (72%) $\gg$ Action Recognition ($\sim$18%)

Limitations of Paper Approach

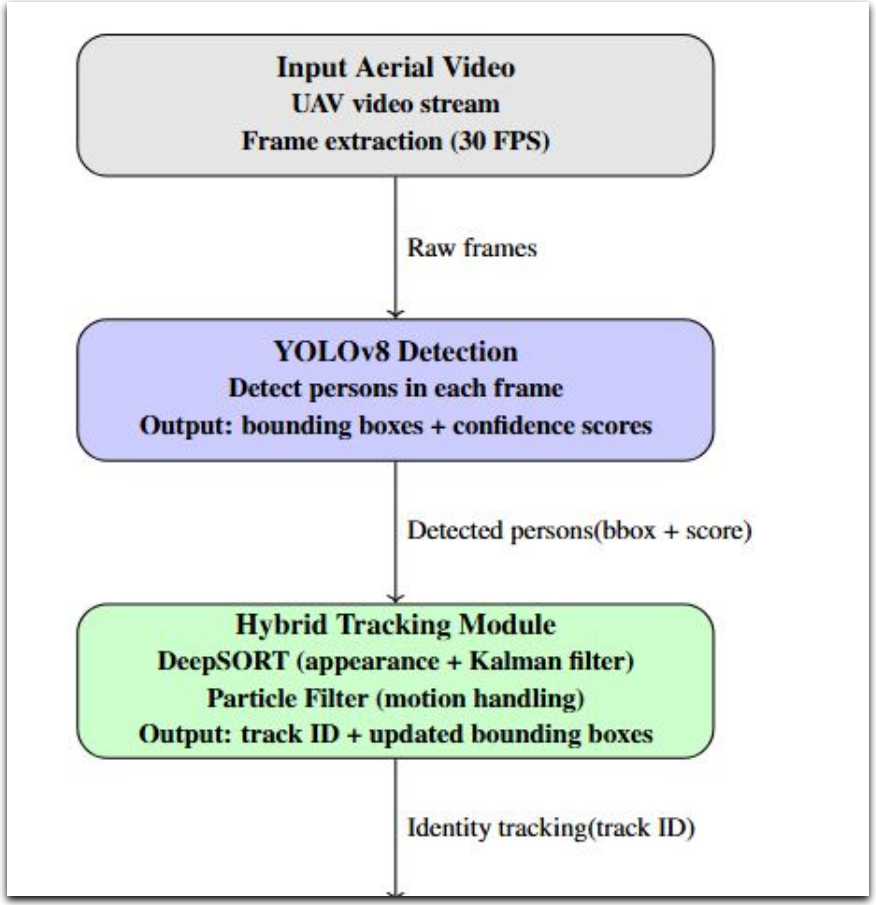
- Frame-level prediction only
  - No temporal modeling
- Cannot handle **multi-label actions**
- No **Tracking** involved
- Poor performance on small-scale humans
- No person-specific understanding
  - Entire frame processed together

Conclusion:

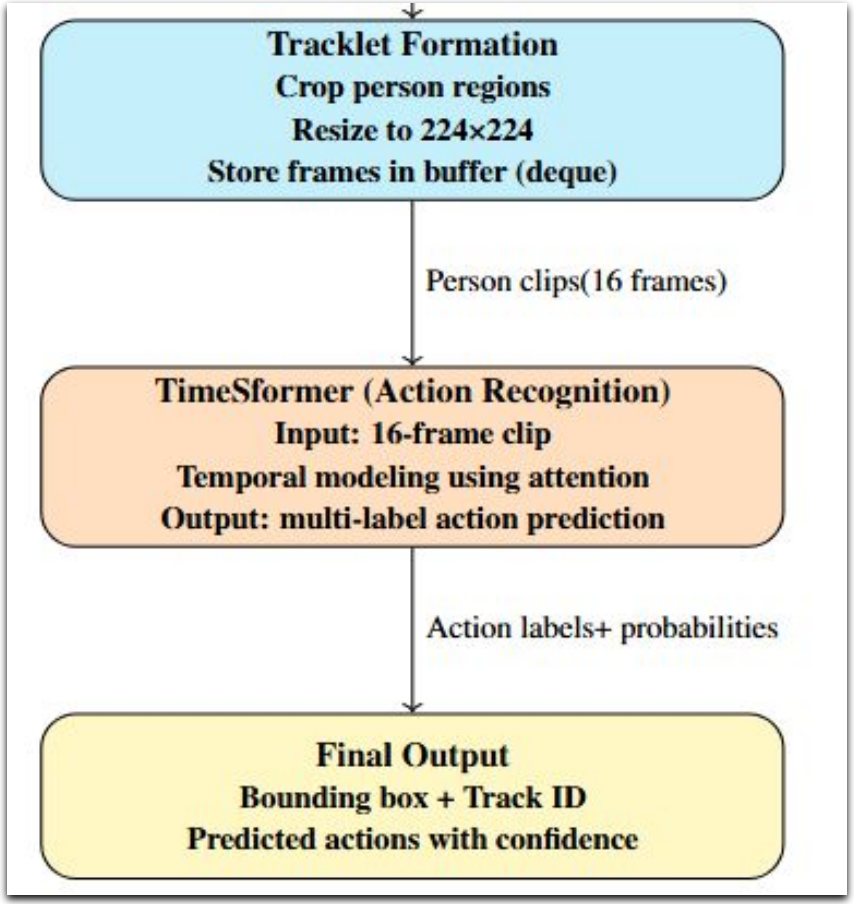
- Requires spatio-temporal and person-centric modeling

Pipeline Overview

Proposed Pipeline



Proposed Pipeline

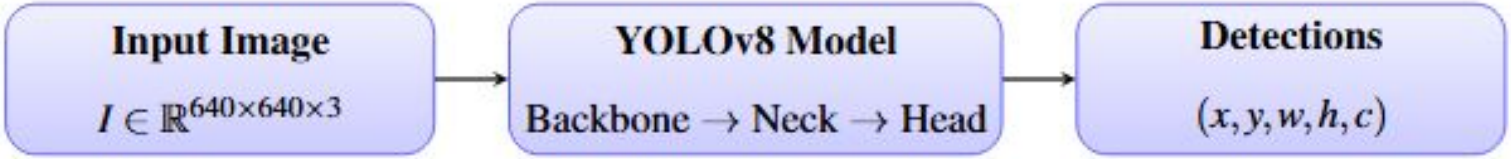


Pipeline Workflow

- **Input Video**
  - UAV video → frame-wise processing (30 FPS)
- **Detection (YOLOv8)**
  - Detect persons → bounding boxes
  - *Answers: Where is the person?*
- **Tracking (Hybrid Tracker)**
  - Assign Track IDs across frames
  - Handles motion, occlusion, and camera movement
  - *Answers: Who is the person?*
- **Tracklet Formation**
  - Crop and store consecutive frames
  - Build 16-frame person-specific clips
- **Action Recognition (TimeSformer)**
  - Predict multi-label actions using temporal modeling
  - *Answers: What is the person doing?*

Proposed Pipeline

Yolo Pipeline:



YOLOv8 in Our Pipeline

- Input frame:

$$I \in \mathbb{R}^{640 \times 640 \times 3}$$

- Output detections:

$$B = \{(x, y, w, h, c)\}$$

- Used for:
  - Detecting persons in each frame
  - Providing bounding boxes for tracking
- **Limitation:**
  - No temporal understanding
  - Cannot recognize actions

Proposed Pipeline

Tracking Pipeline:

How Hybrid Tracking Works

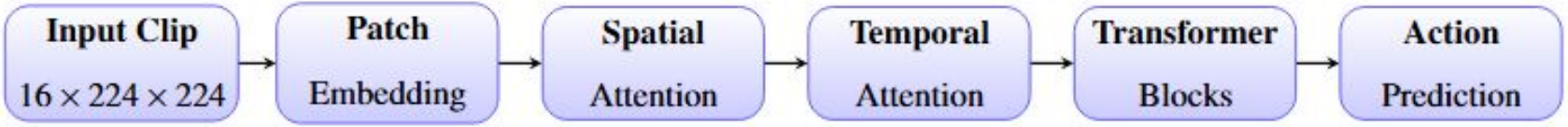
- **Motion Modeling:**
  - Particle Filter predicts possible object states
  - Handles non-linear motion and uncertainty
- **Appearance Matching:**
  - Extract visual features (color / appearance)
  - Helps maintain identity across frames
- **Association:**

$$\text{Similarity} = \alpha\, IoU + \beta\, \text{motion} + \gamma\, \text{appearance}$$

  - Hungarian algorithm assigns detections to tracks
- **Output:**
  - Bounding box + Track ID (per person)

Proposed Pipeline

Timesformer Pipeline:



TimeSformer in Our Pipeline

- Input clip:

$$X \in \mathbb{R}^{16 \times 224 \times 224 \times 3}$$

- Output (multi-label actions):

$$\hat{y} \in \mathbb{R}^{12}$$

- Prediction:

$$y_i = \sigma(\hat{y}_i) > 0.5$$

- Key idea:
  - Spatial attention (within frame)
  - Temporal attention (across frames)
- **Role:**
  - Recognizes actions from person-specific tracklets

Proposed Pipeline

2.5 Comparison of Detection and Action Recognition Approaches

Table 2.1: Comparison of Different Approaches for Aerial Action Recognition

Method	Spatial Info	Temporal Info	Multi-Person Handling
YOLO [10, 4]	Yes	No	NO
Swin Transformer [8]	Yes	No	Yes
TimeSformer [2]	Yes	Yes	Limited
Proposed (YOLO + Tracking + TimeSformer)	Yes	Yes	Yes

Final Output and Key Insight

Final Output per Person:

- Bounding box (location)
- Track ID (identity)
- Predicted actions + confidence scores

Key Insight: Person-Centric Pipeline

- Each individual is processed independently
- Reduces background noise from aerial scenes
- Avoids confusion in multi-person scenarios
- Improves action recognition accuracy

Core Idea:

- Detection $\rightarrow$ **Where**
- Tracking $\rightarrow$ **Who**
- TimeSformer $\rightarrow$ **What**

Algorithm 1: Person-Centric Action Recognition

Video $V = \{I_t\}$ Action predictions $\mathscr{R}$

Initialize $\mathscr{B}, \mathscr{R}$ $\mathscr{B}$ stores temporal sequences per identity, $\mathscr{R}$ stores outputs

each frame I_t

Detection and Tracking:

$$B_t = M_d(I_t), \quad \mathscr{T}_t = \text{Track}(B_t)$$

M_d: detector, $\text{Track}(\cdot)$ performs data association using motion + appearance

each $(id, b) \in \mathscr{T}_t$

Spatial Extraction and Temporal Update:

$$P_t^{(id)} = I_t[b], \quad \mathscr{B}[id] \leftarrow \mathscr{B}[id] \cup \{P_t^{(id)}\}$$

Extract person region and build temporal sequence

$|\mathscr{B}[id]| = \tau$

8 Spatio-Temporal Modeling:

$$X^{(id)} \in \mathbb{R}^{\tau \times H \times W \times 3}, \quad \hat{y}^{(id)} = M_a(X^{(id)})$$

M_a: learns joint spatial-temporal features via attention

9 Multi-label Prediction:

$$\hat{y}^{(id)} \in [0,1]^C$$

Predicts probabilities for C action classes

10 Store and Reset:

$$\mathscr{R} \leftarrow \mathscr{R} \cup \{(id, t, \hat{y}^{(id)})\}, \quad \mathscr{B}[id] \leftarrow \emptyset$$

11 **return** $\mathscr{R}$

Results & Visualizations

YOLOv8 Training Configuration

Model Setup:

- YOLOv8 (medium variant)
- Input Resolution: 640×640
- Batch Size: 32

Training Parameters:

- Optimizer: AdamW
- Learning Rate: 0.002
- Weight Decay: 0.0005
- Scheduler: Cosine decay
- Epochs: 30

Data Augmentation:

- Mosaic, MixUp
- Scaling, Translation, Flipping
- Color augmentation (HSV)

Challenges and Solutions

Challenge	Solution
Small humans	High resolution + mosaic
Overfitting	Weight decay + mixup
Motion + variability	Augmentations
Training stability	AdamW + cosine decay

We use AdamW with cosine decay and strong augmentations such as Mosaic and MixUp to improve generalization, particularly for small objects in aerial data.

YOLOv8 Detection Results

Metric	Value
Precision	0.92
Recall	0.76
mAP@0.5	0.83
mAP@0.5:0.95	0.48

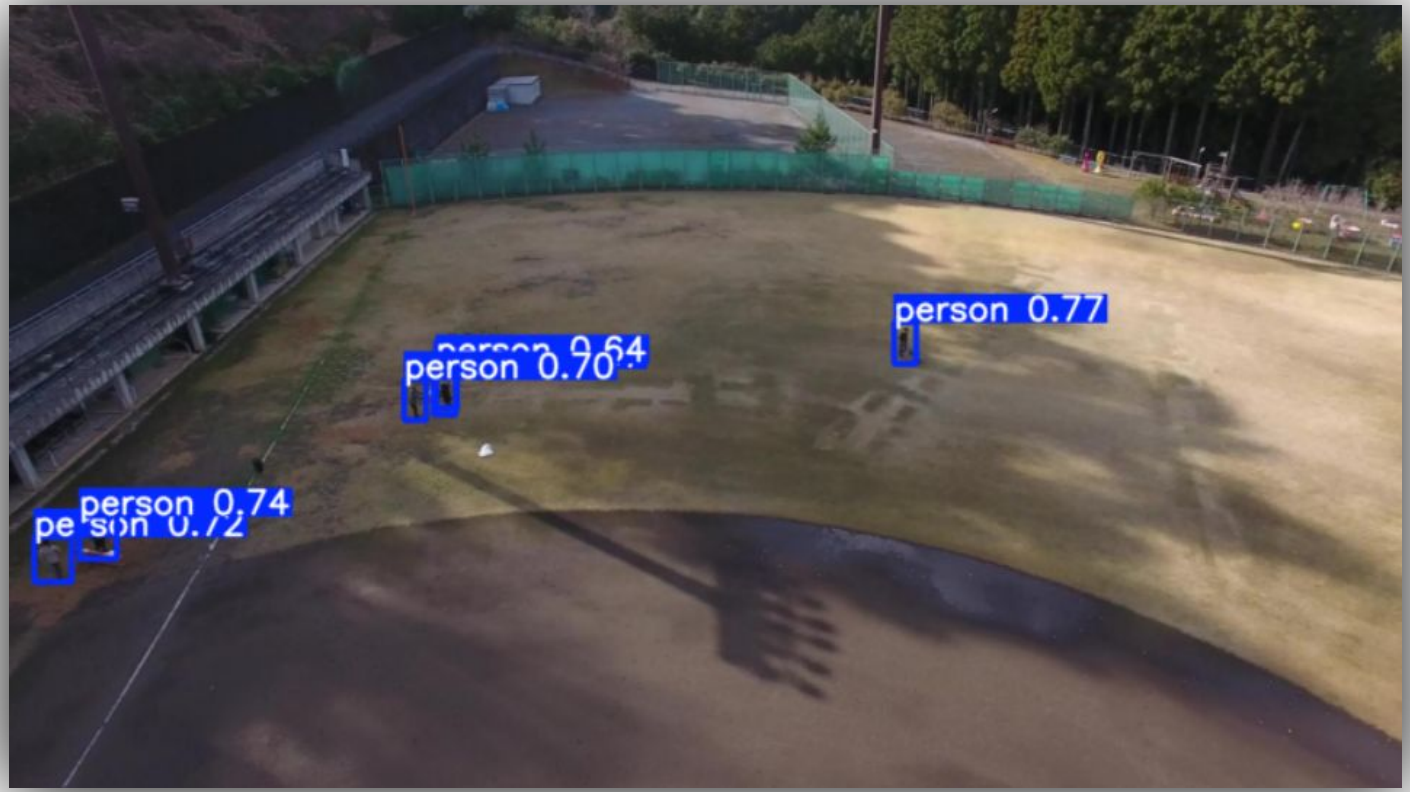
Key Insights:

- High precision → very few false detections
- Strong performance even for small aerial humans
- Runs at **25–30 FPS** → near real-time

Improvement over SSD (Paper):

- SSD: **77.2% mAP** → YOLOv8: **83.2% mAP**

Yolo Visualization



TimeSformer Training Configuration

Input Setup:

- Person-specific tracklets
- Frames per clip: 8
- Resolution: 224×224

Training Parameters:

- Batch Size: 16
- Optimizer: AdamW
- Learning Rate: 1×10^{-4}
- Loss: BCEWithLogitsLoss (multi-label)

Regularization:

- Dropout: 0.3
- Early Stopping (Patience = 15)

TimeSformer Action Recognition Results

Metric	Value
Micro Precision	0.5879
Micro Recall	0.5202
Micro F1-score	0.5520
Macro Precision	0.4376
Macro Recall	0.2956
Macro F1-score	0.3259
mAP@0.5	0.3457

Key Insights:

- Strong performance on **frequent actions** (standing, walking)
- **Micro F1** $>$ **Macro F1** $\rightarrow$ severe class imbalance
- Person-centric tracklets reduce background noise

Challenges:

- Small human size limits fine-grained action recognition
- Subtle actions (reading, calling, drinking) are difficult

Per-Class Action Recognition Performance

Action	Precision	Recall	F1	AP
Calling	0.3636	0.0825	0.1345	0.1243
Carrying	0.4722	0.3789	0.4204	0.4064
Drinking	0.0000	0.0000	0.0000	0.0089
Handshaking	0.3548	0.1196	0.1789	0.1249
Hugging	0.5500	0.1549	0.2418	0.2931
Lying	0.7846	0.2361	0.3630	0.4319
Pushing/Pulling	0.6176	0.5292	0.5700	0.5812
Reading	0.5185	0.1879	0.2759	0.3825
Running	0.0891	0.2222	0.1272	0.0686
Sitting	0.7066	0.5750	0.6340	0.6743
Standing	0.6172	0.6858	0.6497	0.7108
Walking	0.6147	0.6709	0.6415	0.6861

Per-Class Action Recognition Analysis

Performance Across Classes:

- **High performance (frequent actions):**
  - Standing (F1: 0.65), Walking (0.64), Sitting (0.63)
- **Moderate performance:**
  - Carrying (0.42), Lying (0.36), Reading (0.27)
- **Poor performance (rare/subtle actions):**
  - Calling, Handshaking, Running
  - Drinking (F1: 0.00 $\rightarrow$ not detected)

Key Insight:

- Large variation across classes $\rightarrow$ **class imbalance**
- Subtle actions suffer due to **low resolution + weak motion**

Comparison and Key Findings

Performance Comparison:

Method	mAP@0.5 (%)
Baseline (Frame-level)	18.80
TimeSformer (Proposed)	34.57

Key Observations:

- **Temporal modeling improves performance** (+16 mAP)
- **Walking vs Running:** difficult due to top-view similarity
- **Subtle actions:** reading, calling remain challenging
- **Small object size:** limits fine-grained recognition

Conclusion:

- Person-centric tracking improves **stability**
- Still challenging for **fine-grained actions**

TimeSformer Person Level Visualization

TimeSformer Person Level Visualization

Final Pipeline Visualization

Final Pipeline Visualization

Overall Results Summary

Detection (YOLOv8):

- High precision (0.92) $\rightarrow$ very few false positives
- Strong detection performance (mAP@0.5 = 0.83)
- Real-time capability (25–30 FPS)

Action Recognition (TimeSformer):

- Improved performance with temporal modeling (mAP@0.5 = 0.35)
- Gap between Micro and Macro F1 indicates class imbalance $\rightarrow$ strong class imbalance
- Performs well on frequent actions; struggles on subtle ones

Tracking (Hybrid):

- Stable tracking under motion and occlusion
- Reduced ID switches compared to DeepSORT
- Weak long-term re-identification

Overall Insight:

- Person-centric pipeline improves stability and accuracy
- Challenges remain for small objects and fine-grained actions

Why FPS is Low?

Major Bottlenecks:

- **YOLOv8:**
  - Heavy computation on high-resolution aerial frames
- **TimeSformer:**
  - Processes 16-frame clips $\rightarrow$ high GPU load
- Frequent execution of both models reduces throughput

Key Insight:

- Detection + Action Recognition dominate computation
- **Tracking is lightweight** and does not affect FPS

Improving FPS with Smart Inference

Key Idea: Reduce frequency of heavy models

- Run **YOLOv8 + TimeSformer selectively**
  - Execute only on key frames
- Use **Tracking for intermediate frames**
  - Maintains identity in real time
- **Trigger re-detection when needed:**
  - Person exits frame
  - Low tracker confidence
  - New person enters
- **Adaptive strategies:**
  - Increase detection during fast motion
  - Refresh after occlusion
  - Re-run when actions change

How to Increase FPS for Real-Time Performance

```
[ACTION] Frame 24 | Track 1 -> ['Standing', 'Walking']
[ACTION] Frame 32 | Track 0 -> ['Standing', 'Pushing or Pulling']
[ACTION] Frame 32 | Track 1 -> ['Standing', 'Walking']
[ACTION] Frame 40 | Track 0 -> ['Standing', 'Pushing or Pulling']
[ACTION] Frame 56 | Track 0 -> ['Standing', 'Pushing or Pulling']
[ACTION] Frame 56 | Track 1 -> ['Standing', 'Pushing or Pulling']
[ACTION] Frame 72 | Track 0 -> ['Standing', 'Walking']
[ACTION] Frame 72 | Track 1 -> ['Standing', 'Pushing or Pulling']
[ACTION] Frame 80 | Track 0 -> ['Standing', 'Walking']
[ACTION] Frame 80 | Track 1 -> ['Standing', 'Walking']
[ACTION] Frame 80 | Track 2 -> ['Standing', 'Walking']
[ACTION] Frame 88 | Track 0 -> ['Standing', 'Walking']
[ACTION] Frame 88 | Track 2 -> ['Standing', 'Walking']
[ACTION] Frame 96 | Track 2 -> ['Standing', 'Walking']
[ACTION] Frame 104 | Track 0 -> ['Standing', 'Walking']
[ACTION] Frame 104 | Track 1 -> ['Standing', 'Pushing or Pulling']
[ACTION] Frame 104 | Track 2 -> ['Standing', 'Walking']
[ACTION] Frame 112 | Track 0 -> ['Standing', 'Walking']
[ACTION] Frame 112 | Track 1 -> ['Standing', 'Pushing or Pulling']
[ACTION] Frame 112 | Track 2 -> ['Standing', 'Walking']
[ACTION] Frame 120 | Track 0 -> ['Standing', 'Walking']
[ACTION] Frame 120 | Track 1 -> ['Standing', 'Pushing or Pulling']
...
[ACTION] Frame 296 | Track 4 -> ['Standing', 'Walking']
[ACTION] Frame 296 | Track 5 -> ['Standing', 'Walking']
[INFO] Processed 300 frames in 16.73s => 17.93 FPS
```

Conclusion and Limitations

Conclusion

- Person-centric pipeline combining detection, tracking, and temporal modeling
- YOLOv8 achieves strong detection performance:
  - mAP@0.5 = 0.83, Precision = 0.92
- Hybrid tracking improves identity consistency under occlusion
- TimeSformer enables multi-label action recognition:
  - Micro F1 = 0.552, mAP = 0.3457
- Frequent actions detected reliably; fine-grained actions remain challenging

Limitations

- Performance drops for small or heavily occluded objects
- High computational cost limits real-time deployment
- Tracking errors propagate to action recognition
- Class imbalance affects rare action detection

Thank you!